

Networks, signals, and parameters

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Developer guide

Networks

Network is the mutable authoring object. Compilation turns it into immutable graph data. Every graph has a name and timestep.

```
net = snn.Network("example", dt=0.1 * snn.ms)
```

Names must be unique and contain no whitespace. They become stable identifiers in graphs, recordings, and checkpoints.

Signals and inputs

Signals carry an identifier, shape, physical unit, and signal type. Time-varying inputs use the canonical (time, batch, channels) axis order.

```
image = net.input(  
    "image",  
    shape=("time", "batch", 784),  
    signal_type="spikes",  
    unit="spike",  
)
```

This declares what the graph accepts. It does not decide whether the spikes came from MNIST, SHD, a Poisson encoder, or a recorded event stream.

Populations

A population contains equally configured neurons and exposes named ports such as **spikes**, **voltage**, **excitatory**, and **inhibitory**.

```

excitatory = net.population(
    "E",
    size=80,
    neuron=snn.COBA_LIF(
        capacitance_nf=1.0,
        leak_us=0.05,
        resting_mv=-65.0,
        threshold_mv=-50.0,
        reset_mv=-65.0,
        refractory_steps=12,
    ),
)

```

The authoring API describes COBA-LIF, LIF, and non-spiking leaky-integrator populations. The graph executor currently supports COBA-LIF and leaky integrators. General current-based LIF execution is not implemented.

Parameters

Parameters have stable names, shapes, units, initializers, and optional constraints. Projections normally create their own dense weight parameter, but a named `ParameterRef` may be supplied explicitly.

`Normal`, `Constant`, and `NonNegative` are currently available. More elaborate initialization and structural sparsity need explicit reusable representations rather than hidden tensor manipulation.

API reference

Network

```

Network(name: str, *, dt: Quantity = 0.1 * ms)

```

`name` becomes the graph identifier. `dt` is a time `Quantity`. Names claimed inside one network must be non-empty, unique, and contain no whitespace.

Network.input

```

net.input(name, *, shape, signal_type, unit="1") -> Signal

```

`shape` is a tuple of integers and symbolic axes. Time-varying dense inputs begin with ("time", "batch", ...). The returned signal identifier is `<name>.value`.

Network.population

```

net.population(name, *, size, neuron, spiking=True) -> Population

```

`size` must be positive. A spiking population exposes `.spikes` and `.voltage`; a non-spiking population exposes `.voltage`. Target ports are `.excitatory`, `.inhibitory`, and `.modulatory`.

Network.parameter and Network.constant

```
net.parameter(name, *, shape, initializer, unit="1", constraint=None) ->
ParameterRef
net.constant(name, value, *, unit="1") -> str
```

Parameter constructors are `Normal(mean, std)` and `Constant(value)`. `NonNegative()` is the implemented constraint. Neuron constructors are `COBA_LIF(**values)`, `LIF(**values)`, and `LeakyIntegrator(**values)`. Unit helpers are `ms`, `mV`, `nS`, and `Hz`.

Core return types

`Signal` exposes `id`, `shape`, `unit`, `signal_type`, `owner`, and `port`. `Population` exposes `id`, `size`, `neuron`, `spiking`, `group`, and its signal or target-port properties. `ParameterRef` contains the stable parameter id.

Next: Components, projections, and delays